

ID : 14661

Name : Rida Bilal

"Assignment : 01"

Form a linear regression equation from the following data to find how much the revenue will be at \$8000 spent.

Using standard form of a linear regression equation

$$y = a + bx \rightarrow (i)$$

Amount spent In Ads	1	2	3	4	5
Sales (1000 \$)	2	2.8	3.6	4.5	5.1

In eq (i)

y is the predicted value (sales)

x is the Independent variable (add spent)

a is intercept

b is slope

x	y	xy	x ²
1	2	2	1
2	2.8	5.6	4
3	3.6	10.8	9
4	4.5	18	16
5	5.1	25.5	25

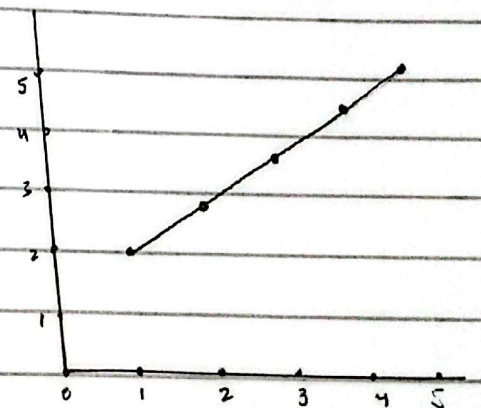
$$\sum x = 15$$

$$\sum y = 18$$

$$\sum xy = 61.9$$

$$\sum x^2 = 55$$

$$n = 5$$



Slope = ?

$$b = \frac{n (\sum xy) - (\sum x)(\sum y)}{n (\sum x^2) - (\sum x)^2}$$

$$b = \frac{5(61.9) - 15(18)}{5(55) - 15^2} \Rightarrow \frac{309.5 - 270}{275 - 225} \Rightarrow \frac{39.5}{50}$$

$$b = 0.79$$

Intercept = ?

$$a = \frac{\sum y - b \sum x}{n}$$

$$a = \frac{18 - 0.79 \times 15}{5} \Rightarrow \frac{6.15}{5}$$

$$a = 1.23$$

When 8000 \$ spent on ads $x = 8$

$$y = a + bx$$

$$y = 1.23 + 0.79 \times 8$$

$$y = 1.23 + 6.32$$

$$y = 7.55$$

Predicted sales are 7550 \$